

Documenting Register Data for Research Purposes



REKISTERITUTKIMUKSEN TUKIKESKUS
INFORMATIONSCENTER FÖR REGISTERFORSKNING
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Finnish administrative registers

- several comprehensive national registers
- contain data on **individuals, families, housing, enterprises**
- compiled and maintained for administrative or statistical purposes, e.g.
 - ***Social Insurance Institution (KELA)*** pays social benefits to Finnish citizens and needs registers with data on individuals for this purpose
 - ***National Institute for Health and Welfare (THL)*** is a register and statistical authority and collects individual based data from local registers for compiling statistics and for research purposes
 - ***Statistics Finland (Tilastokeskus)*** is a statistical authority and collects data on individuals mostly from administrative registers for compilation of different statistics

Secondary usage of administrative registers

- Production of official statistics is almost totally based on registers in Finland
 - *the vast majority of data (over 95%) come from administrative registers*
 - *the population and housing census has been based totally on register sources since 1990*
 - *Handbook: Use of Registers and Administrative Data Sources for Statistical Purposes – Best Practices of Statistics Finland*
- Register-based research
 - *20 % of doctoral thesis' within medicine in Finland include data from national registers*

Prerequisites for register-based research

- Common personal identification number in all registers
 - first used in 1964 (between 1964-1970 two different systems)
 - since 1971 a digital population register
 - all Finns have a PIN

→ data from different registers can be linked by PIN e.g. for research purposes
- Legislation that allows the use of confidential personal data for scientific research
- Comprehensive, well documented registers

Current state of register metadata in Finland

- Description of data file for files containing personal data
 - Required by law (Personal Data Act)
- Handbooks
 - Statistics Finland Handbook on population census
 - http://www.stat.fi/tk/he/vaestolaskenta/vaestolask_opas2000_en.pdf
 - Guidebook for collecting and sending data to the hospital discharge register
 - Compiling data from hospital registers to national register
 - The IT-registers of the Social Insurance institute and data included into them (pdf-file, 260 pages)
- Definition of concepts, national classifications (eg. classification of education) and international classification standards (eg. ICD10, ISCO) on internet
- Technical specifications of data file
 - Only for internal use at the register keepers
- No metadata standards in use

Description of personal data file (section 10 of Personal Data Act)

- Content
 - Registrar
 - Person responsible of the register
 - Name of register
 - Purpose of the register
 - Contents of the register
 - Description of group or groups of data subjects and the data relating to them
 - Regular sources of information
 - Regular disclosure of information
 - Register protection principles
- The Register description should be kept available

Example of list of data items in the Finnish population census data file

- **Identity code**
- **Age**
Age in years at 31 Dec 2000
- **Sex**
 - male
 - female
- **Marital status**
 - unmarried
 - married
 - widowed
 - divorced
- **Language**
- **Nationality**
- **Religion**

Main type of activity (LF)

Labour force

- employed
- unemployed

Outside labour force

- 0-14-year-old
- student
- pensioner
- conscript, conscientious objector
- other

Occupational status

- wage earner
- self-employed

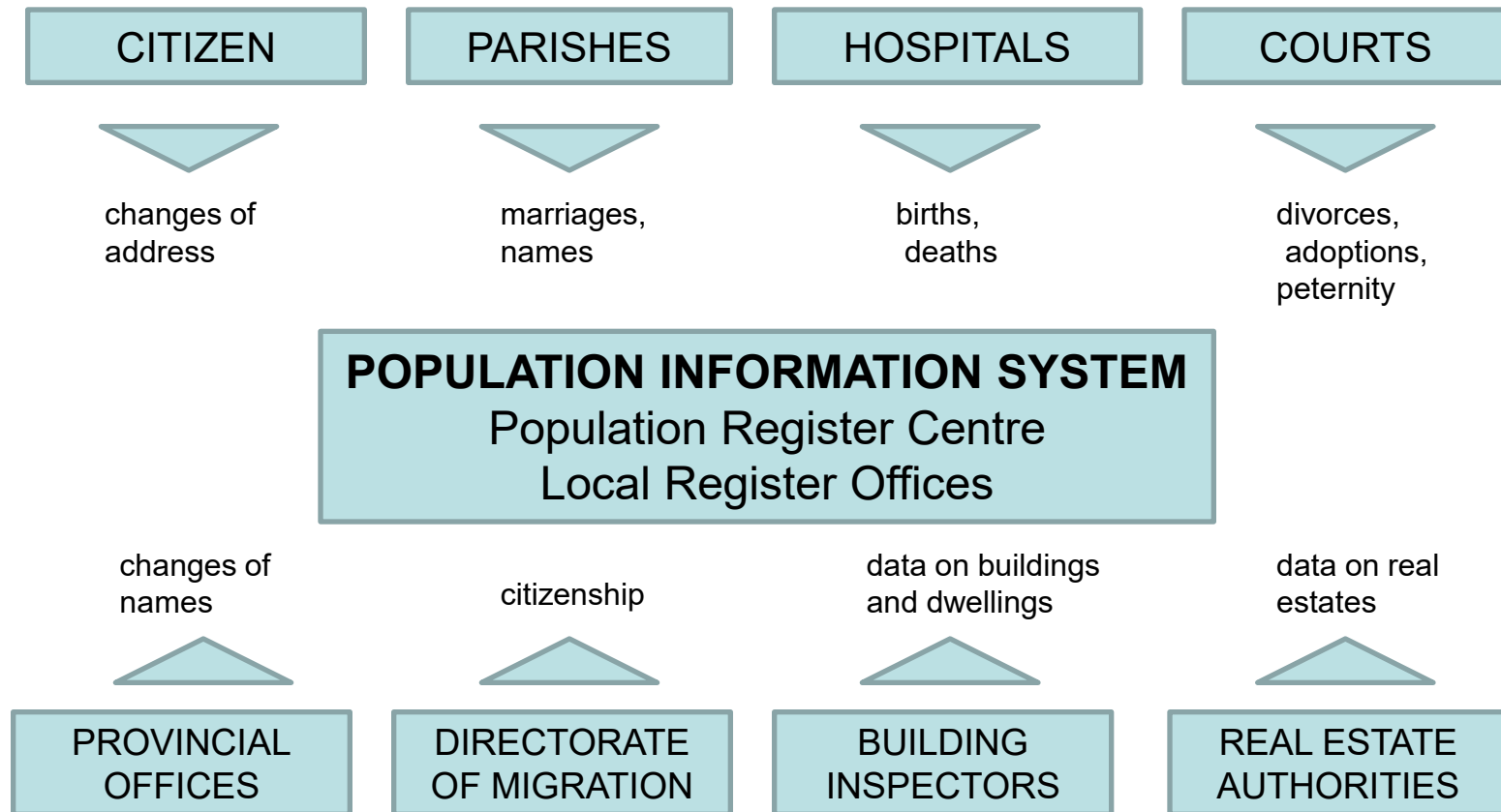
Occupation

Code according to Statistics

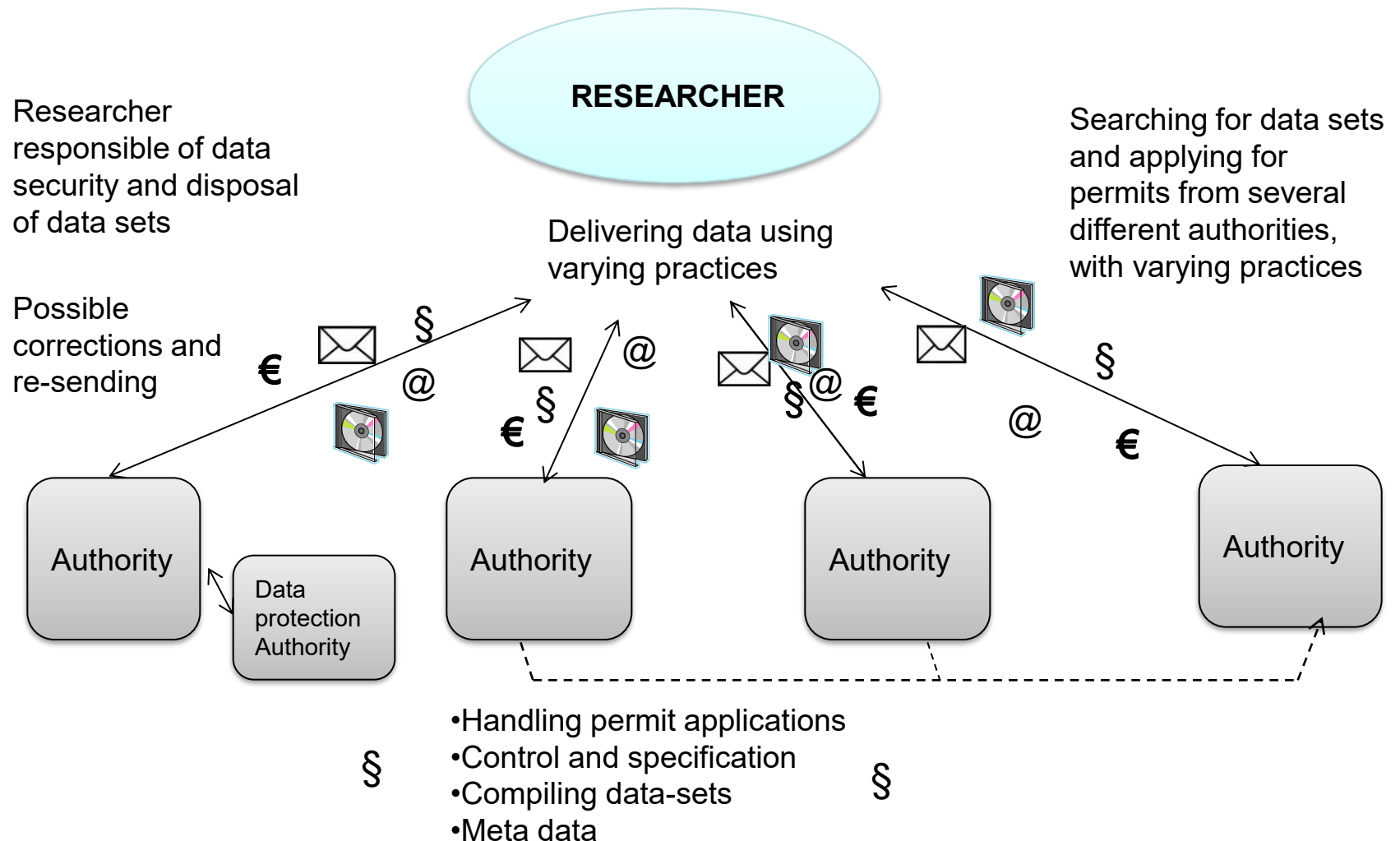
Finland's 1997 and

2001 classifications of occupations.

Authorities responsible for updating the population information system



Present process for obtaining register data for research



Researcher

MIDRAS-remote access system

Services that require permit

- Remote desktop for analysing data (programs and tools)
- Separated server space for data and metadata
- Output service for results, Input service for researcher's data

Services that require registration

- Centralized digital permit application service

Public services

- Metadata catalogue
- Helpdesk for research and tuition

Interface service for data and meta data,

Administration services for user rights

Pseudonymization

Organiza
-tion A

Organiza
-tion B

Organiza
-tion C

Organiza
-tion D

Organiza
-tion E

- Commonly agreed metadata standards
- Data warehouse
- Archive of multiple user files

Metadata requirement specifications

- Four levels
 - Description of the metadata
 - Eg, standards, date of change, version, source, language, openness
 - Description of the data file or register
 - Eg. Name, organization, contact information, subject unit, coverage, sample, size, variable quantity, abstract, time, frequency, changes, processing, format, data source, version date, version, id, primary use
 - Description of the variables
 - Variable name/label, definition, type, code, version date, processing, missing value, source, date, coverage, right to use, safety class, obligatoriness,
 - Description of classifications
 - Code name, code value, code definition, reference, version date

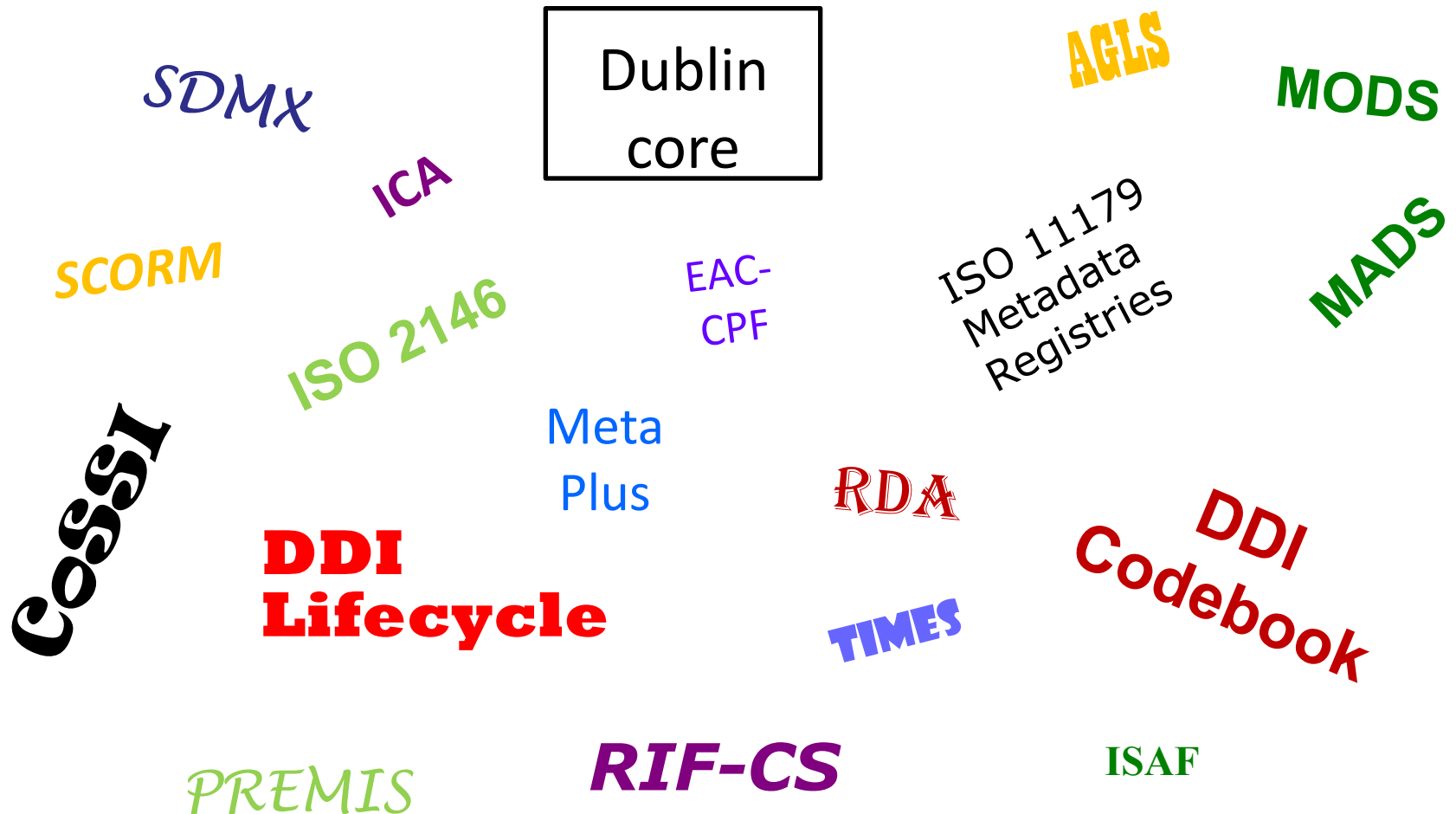
Metadata requirement specifications (listing)

Number	Metadata	Explanation	Example	Importance
M1	Variable	Name of variable in the register	HTIKA	Must (A)
M2	Definition	Description of what has been stored in the variable	The parent's age at the beginning of the year	Must
M3	Type	Possible values of variable or presentation	Number between 0-120 / date as yymmdd	Must
....				

To take into consideration for registers

- Collected over a long time span
 - Content changes over time
 - Classification changes
 - Collection mode changes
 - Laws / grounds for collecting data changes
- Many separate parties involved
 - Collection on local level into local registers
 - Compilation into national registers
 - (Processing into statistical registers)
 - Extraction of research data file
- Same variables and classifications in different registers
- Stored either in relational databases or as sequential files

Search for metadata standard, -model,-format



Current metadata work in Finland

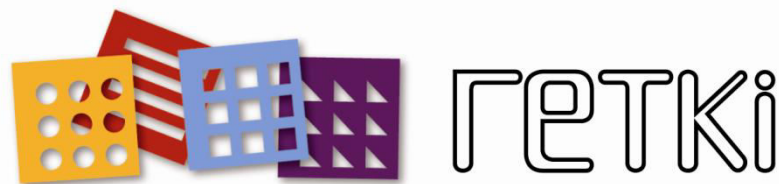
- Statistics Finland
 - CoSSI (Common Structure of Statistical Information)
 - Developed and used within Statistics Finland
 - Tools: Metadata editor
- National Archives of Finland
 - Project SÄHKE3
 - Project for developing electronic information management and long-term preservation of register data and databases
 - SÄHKE3-norm to be implemented in 2013-2014
- Finnish Social Science Data Archive
 - Presently using DDI -C, studying DDI -L
- And many others

Challenges

- Find a suitable metadata standard(s) for registers
- Reach consensus
- Implement the standard at all levels of data collection and processing
 - What's in it for the register keepers?
 - Shouldn't be too much work!
 - National standards
 - JUHTA (Advisory committee on Information Management in Public Authorities)
 - JHS (Public Administration Recommendation)
 - Plans to appoint a group for register recommendations
 - SÄHKE3

Thank you!

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